

Personal Profile

Double Master's student (Sorbonne/USI) in Applied Mathematics with peer-reviewed publications at NeurIPS and COLING in continual learning and explainable deep learning. Coursework in parallel programming on CPUs and GPUs, Scientific Machine Learning and Numerical Linear Algebra. Currently, developing tabular machine learning methods at Institut Pasteur and Paris Brain Institute. Looking for a PhD in Machine Learning.

Education

- 2024–Ongoing **Sorbonne Université**,
Mathematics and Computer Science, Double Degree Master of Science, Paris, France.
○ Coursework: Numerical Linear Algebra, Parallel GPU/CPU Programming, Numerical Analysis of Neural Networks, Floating-Point Arithmetics
- 2024–Ongoing **Università della Svizzera italiana (USI)**,
Computational Science, Double Degree Master of Science, Lugano, Switzerland.
○ GPA: 9.33/10 (Swiss system)
○ Coursework: Scientific Machine Learning, Graph Deep Learning, Stochastic Methods, Computer Architectures, High-Performance Computing
- 2019–2023 **Osnabrück University**,
Cognitive Science, Bachelor of Science, Osnabrück, Germany.
○ GPA: 1.3/6.0 (German system, best: 1.0)
○ Specialisation in Mathematics, Computer Science, Machine Learning, and Computational Neuroscience
○ Bachelor Thesis on Interpretability of LLMs with grade A, published at NeurIPS'23 workshop & COLING'24.

Publications

Peer-Reviewed Conference Publications

- COLING'24**, **Ackermann, L.**, & Ohmer, X. I. (2024). On the Relationship between Skill Neurons and Robustness in Prompt Tuning. In Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (pp. 12403-12415) [paper][code] (*travel grant awarded*)
Turin, Italy

Peer-Reviewed Conference Workshop Publications

- NeurIPS'24**, **Ackermann, L.**, & Nguyen, V.-L. (2024). On the transferability of parameter-efficient continual learning for vision transformers. In NeurIPS 2024 Workshop on Fine-Tuning in Modern Machine Learning: Principles and Scalability. [paper]
Vancouver, Canada
- NeurIPS'23**, **Ackermann, L.**, & Ohmer, X. I. (2023). On the Relationship between Skill Neurons and Robustness in Prompt Tuning. In NeurIPS 2023 Workshop on Robustness of Few-shot and Zero-shot Learning in Large Foundation Models (RO-FoMo) [paper][code] (*travel grant awarded*)
New Orleans, USA

Software Publications

- Zenodo/CRAN** **Ackermann L.**, Zhuojun G. & Kjell O.N.E. (2024). An R package for visualizing text in topics. DOI:zenodo.org/records/11165378, [website][code] (*42k downloads*)
- PsyArXiv** Eijsbroek, V. C., **Ackermann, L.**, Nilsson, A., Gu, Z., Wiebel, C., Ganesan, A. V., Kjell, O. N. E. (2025, July 23). Multiple Methods for Visualizing Human Language: A Tutorial for Social and Behavioural Scientists. [paper]

Experience

- 03/2026– **Master Thesis Intern** [code],
08/2026 INSTITUT PASTEUR & PARIS BRAIN INSTITUTE, Paris, France.
◦ Advancing capabilities of tabular machine learning methods (e.g. TabPFNv2)
- 04/2024– **Machine Learning Working Student** [code],
09/2024 HASSO-PLATTNER-INSTITUT, Potsdam, Germany.
◦ Developed an evaluation pipeline with PyTorch that integrates six Deep Learning-based data synthesizers.
◦ Utilized Docker, Slurm, and Nvidia Pyxis to run the pipeline on the High-Performance Computer.
- 06/2024– **Machine Learning Intern** [paper],
08/2024 CYBER INFORMATION SECURITY LAB, CCU, Chiayi, Taiwan.
◦ Conducted research project on Deepfake Detection, published at NeurIPS'24 workshop
◦ Performed the first benchmarking of six continual learning methods on advanced Vision Transformers.
- 10/2023– **Machine Learning Intern** [website][code],
01/2024 LUND UNIVERSITY & ABLEMIN, Lund, Sweden.
◦ Developed the R package "topics" (28k+ downloads), providing a scientific topic modeling pipeline with statistical evaluations and visualizations.

Projects

- 01/2026– **Multi-Node GPU/CPU optimization for n-body simulation** [report][code],
02/2026 SORBONNE UNIVERSITÉ, Paris, France.
◦ Scaling vectorized n-body solver across nodes with multiple GPUs and multiple CPUs
◦ Software-hardware scaling analysis for MIPP, MPI, CUDA and OpenMP based on hardware specifications
- 01/2026– **Parallelizing Domain Decomposition for the 2D Helmholtz** [report][code],
02/2026 SORBONNE UNIVERSITÉ, Paris, France.
◦ Scaling MPI implementation of parallel Interface problem solver across multiple nodes for Helmholtz equation
- 03/2025– **FEM vs. DeepRitz model for reaction-diffusion equation** [report][code],
06/2025 UNIVERSITÀ DELLA SVIZZERA ITALIANA (USI), Lugano, Switzerland.
◦ Optimized DeepRitz method for a dynamic reaction diffusion equation modeling the heart tissue
◦ Improved the numerical solver by using different collocation point sampling and enforcing initial conditions
- 03/2025– **Postprocessing of MeteoSwiss' wind prediction with Graph Deep Learning**, [report][code],
06/2025 UNIVERSITÀ DELLA SVIZZERA ITALIANA (USI), Lugano, Switzerland.
◦ Improved the postprocessing for MeteoSwiss official wind speed forecasts in Switzerland
◦ Designed a statistical analysis between Graph-based and non-graph-based models

Scholarships

- 06/2024– **EUMaster4HPC**.
Ongoing EU HPC Excellence Program to educate the next talented generation of High-Performance Computing experts.
- 04/2020– **Scholarship of Ernst-Ludwig-Ehrlich Studienwerk (ELES)**.
Ongoing ELES is the German scholarship foundation for gifted students of the Jewish community in Germany.

Skills

- Programming Python, C++, C, SQL, R, MATLAB, Bash
- Frameworks Keras, PyTorch, Tensorflow
- HPC Tools Slurm, OpenMP, CUDA, MPI, MIPP
- Development Git, Docker, Linux
- Languages German (Native), English/French (Advanced), Swedish (Intermediate), Chinese (Beginner)

Extra Curriculars

- 03/2024– **AI Education Volunteer**, *KI macht Schule*, Berlin.
Ongoing Educating high school students about artificial intelligence.
- 03/2021– **Mathematics Tutor Volunteer**, *Lern-Fair*, Remote.
04/2023 Tutored disadvantaged students in mathematics for final exams.